Chapter 14
Enhanced Two-Step Floating Catchment Area (E2SFCA) Method for Measuring Spatial Accessibility to Primary Healthcare in HD Kote, Mysore (India)

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Abstract Accessibility to the health centre is poorly understood in most of the developing countries. Regardless of development, it is a challenge to provide healthcare services to the entire population. The study aims to measure the spatial accessibility of the rural population in HD Kotetaluk of Mysore district using the gravity model (2SFCA). The spatial accessibility has been measured by using GIS-based network analysis. The health centre to population ratios (R_j) for villages have been calculated by dividing the number of serving health centres by the population of that village. The reciprocal of R_j of a particular health centre is calculated by summing up reciprocals of R_j values of all villages within its catchment area; keeping in view the frequency of health centres serving a particular village. The Accessibility index (A_i) value for a particular catchment area is calculated by summing up the R_j values of serving health centers within a catchment area. The paper also aims at assessing the impact of socio-economic factors that influence the accessibility and utilization of health services in the study area. The study establishes that health centres follow a clustered pattern with around 80% of the population having access to the nearest health facility. The study is helpful for the policymakers to identify the inaccessible areas and also demarcate the service area of healthcare centres. Such steps will be helpful to improve the accessibility to healthcare in terms of improving the number of facilities and also improving the road network in an area.

Keywords Spatial accessibility · Gravity model · 2SFCA · Network analysis · Service area · Mysore

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14.1 Introduction

Hospitals are the most important places to treat the health-related issues of people and raise the quality of good living (Halder et al. 2020). The measurement of access to health services is important for the health status of the population and its variability for the effective allocation of national health resources (Bryant and Delamater 2019). Accessibility can be measured based on financial costs, infrastructure facilities, quality of medical care, acceptability and geographic accessibility (Mansour 2016). The two-step floating catchment Area (2SFCA), introduced by Radke and Mu (2000) repeats the floating process twice (once at the supply sites and once at the demand sites). The total number of beds, the total number of doctors, transport supply, population density and others as assessment factors for accessibility to healthcare facilities (Higgs and Gould 2001). A growing priority for health service planners is to minimize the inequalities in spatial access to health services. Therefore, there is an urgent need to identify the human population habitations which do not have access to healthcare facilities, but efforts to quantify such disparities in spatial accessibility have been hampered by a lack of satisfactory measurements and methods (Ngui and Apparicio 2011). Poor access to healthcare facilities can take a toll on people suffering from basic health issues resulting in serious consequences and therefore developing more complicated problems (Brabyn and Gower 2003) with the passage of time.

Five commonly used spatial accessibility criteria are: (1) the distance from the nearest service, (2) the number of services over a given length of time, (3) the average distance from all services, (4) the average distance from the nearest service, and (5) the gravity model (Apparicio et al. 2008). The key drawback of the distance approach to the nearest service is that, without taking account of availability, it captures only the proximity between the population and the places of service (McGrail and Humphreys 2009). The 2SFCA method is a special case for quantifying a gravitational model on the GIS platform that makes it easier for authorities to identify areas of accessibility and inaccessibility in a region, and this would help to take appropriate measures for infrastructure development to improve the road network, as well as to identify optimal locations for new health centres in areas which do not exist (Kanuganti et al. 2016). The two-step floating catchment area (2SFCA) is a popular technique, especially in healthcare systems for measuring spatial accessibility. In its characterization of spatial accessibility, 2SFCA is based on a potential model that incorporates the interaction between supply, potential demand and travel costs. There are two fundamental factors among the many, influencing access to health services which are: the supply of doctors and the demand of the population. Both are spatially distributed, but it is rare for their distributions to be perfectly combined. It has become a significant measure of spatial accessibility over the last decade, especially in its application of access to basic health care. Several recent improvements have been developed to the original 2SFCA system, which typically explains the decrease in the distance in a catchment or enables the use of various measurements of variable catchment size (McGrail 2012). The Floating Catchment Area (FCA) is a system focused on gravity

specifying the physicians' travel time threshold to the service area (Luo and Qi 2009). Numerous studies have employed the geographic information system (GIS) techniques for facility locations (Pramanik 2016).

Access to health care services is affected by the location, as health professionals are fully aware regarding the supply and demand areas (from the location and population point of view) and also the healthcare facilities which are not fairly distributed (either from the population or the health professionals/human resource point of view). In far-flung rural and deprived urban areas, the shortage of doctors and other related human resources has been especially serious (Rosenblatt and Lishner 1991). The ratio of population to health workers is the key factor in estimating spatial trends. One of the recent developments, such as the use of multiple health centre buffer study, is to resolve the population limitation of the region-based doctors (Luo 2004). The Floating Catchment Area (FCA) approach can be helpful if complete details (address, road accessibility, distance, etc.) of a person are available to resolve the presumption of the cross-boundary approach (Horev et al. 2004). In a two-step technique, a floating catchment area can be expanded. The catchment area of health facilities is measured in the first approach based on travel time and the catchment around each demand centre is measured in the second approach based on travel time (Radke and Mu 2000). It is a special case of the gravity model and also intuitive to interpret as it simply uses a special form of relationship between physicians and a population (Kanuganti et al. 2016). In recent times many improvements have been observed to the original two-step floating catchment area method which generally either accounts for distance-decay within a catchment or enables the usage of variable catchment sizes, therefore, it is necessary to use both demand and supply functions to measure the accessibility to healthcare facilities in a region or an area (McGrail 2012). For measuring the healthcare accessibility to lower-order facilities (PHCs), the higher-order healthcare facilities (Taluk/District Hospital and Community Health Centres) has to take into consideration because higher-order health centres provide all health facilities which are available in lower-order health centres (Tali et al. 2017a).

14.2 Study Area

Heggadadevanakote (HD Kote) taluk is one of the seven taluks (sub-district) of Mysore district. The total geographical area of the taluk is 1618 sq. Km and accounts for about 8.7% population of the district. The study area has 16 primary health centres, one Community Health Centre (CHC), and one Taluk hospital. The study area has an average elevation of 895.378 m above sea level with a maximum and minimum elevation of 623 and 1145 respectively. A significant proportion of residents have to travel a long distance to reach health care services, especially in the study area's south and south-western regions (Minutha et al. 2014). The population of the study area is 267306 and the PHC-population ratio is 14850 which is within

the prescribed criterion as fixed by National Health Mission (NHM) but the location of these facilities does not match up with the distribution of population which has resulted in the variations of accessibility in the study area.

14.3 Database and Methodology

14.3.1 Database

The study is based on primary as well as secondary sources of data. The primary data was collected by a patient-exit survey. The primary survey has been conducted to analyse the socio-economic problems related to accessibility through questionnaires at the health centre level. With the prior written consent of the patients who agreed to take part in the survey after explaining to them the nature and purpose of the study, however, they were clearly briefed about all the procedures used in the study. 200 patients have been interviewed out of that 160 at Primary health centres PHC's (10 at each PHC) and 20 at community health centres (CHC) and the remaining 20 at Taluk Hospital. The patient-exit survey has been conducted to measure the availability, affordability, accessibility and quality of service availability at these facilities. Quality of health services includes the 19 types of quality services that were asked to the respondents to analyse the quality of healthcare facilities available in the health centres and they were asked to give a rating between poor, fair, good, and excellent. Additionally, the location of existing health centres was collected through a GPS survey. The population data and village boundary maps were obtained from the census department. Further, this collected information was integrated with ArcMap and the spatial database was prepared. Network analysis tool in ArcMap has been used to measure the physical accessibility in terms of distance to and from health care centres. The elevation data of the study area has been generated from SRTM DEM of 90 m spatial resolution (Fayaz et al. 2020) downloaded from USGS and the same has been integrated with the road network. The drive time was calculated based on the expected vehicular traffic on the respective highways, road hierarchy, land use in the area, road width, condition of the road, traffic bottlenecks, and other factors, and does not include traffic signals because emergency vehicles are permitted to travel through traffic signal points. The drive time is only applicable in regular traffic conditions; it does not exist in cases where there are sudden traffic delays or other obstacles, such as tree falls, road collisions, road blockades, and so on. The acceptable distance for the service area was fixed by consulting the patients of the villages at seven PHCs while utilizing the health centres for basic healthcare. Therefore, the service area from health centres has been measured up to a distance of 9 km which is convenient for patients who feel enough comfortable to travel the same distance.

14.3.2 Methodology

GIS network analysis tool (service area and closest facility) was used to execute the 2SFCA method. Two-step floating catchment area method accounts area twice based on the demand of the population i.e. Health centre to population ratio and supply from the facility i.e. accessibility index of the population.

i. *Health centre Population ratio (R_j)*

$$R_j = \frac{S_j}{\sum_i P_i} \tag{14.1}$$

where:
S_j is the number of full-time equivalent services at location *j*,
P_i is the number of residents at population location *i* and
R_j is the health centre to population ratio for service *j*.

ii. *Accessibility Index of Population (A_i)*

$$A_i = \sum_j R_j \tag{14.2}$$

where:
A_i is the accessibility for population at location *i*.

iii. *Reciprocal of R_j*

$$\frac{1}{Rj} = \sum_{i=1}^{n} \frac{1}{R_{ij}} \tag{14.3}$$

where $\frac{1}{Rj}$ is the *Reciprocal of* R_j

14.3.3 Global Moran's-I

In the present study, Global Moran's-I was applied to understand the spatial distribution of health facilities by measuring spatial autocorrelation using feature locations and feature values simultaneously. The formula applied for the calculation of the spatial patterns of health centres is as Wong (2005).

$$I = \frac{n \sum_{i=1}^{n} \sum_{j=1}^{n} W_{ij} \left(X_i - \dot{X}\right)\left(X_j - \dot{X}\right)}{\sum_{i=1}^{n} \sum_{j=1}^{n} W_{ij} \sum_{i=1}^{n} \left(X_i - X\,\dot{X}\right)} \tag{14.4}$$

where n is the number of features, X_i is the attribute. W_{ij} is the assigned weight to each of the village (polygon) based on their neighbour definition.

14.4 Results

14.4.1 Spatial Distribution of Healthcare Facilities

The calculated Moran's-I, in this case, is 0.252274 as shown in Fig. 14.2, which indicates the clustered nature of healthcare facilities in the study area. The standardized score which is 3.211920, indicates the clustered pattern of health facilities over 243 villages and is statistically significant at 99% (0.001319) confidence level.

14.4.2 Spatial Accessibility of Healthcare Facilities

The service area of healthcare facilities and the closest facility of villages have been generated based on travel distance within the impedance cut-off of 9 km (acceptable distance). Based on this 80.52% of the population of the study area is accessible and the remaining 19.48% is inaccessible to the healthcare system of the area.

The distribution of healthcare centres, as shown in Figs. 14.1 and 14.2, it is obvious that healthcare facilities have cluster distribution patterns as shown by Directional Distribution (Standard Deviational Ellipse) and Global Moran's-I. Due to this, the service areas of the healthcare centres are overlapping i.e. one village is being served by more than one healthcare centre. In this case, R_j for each village was calculated by dividing the number of serving healthcare centres by the population of that village. The numerator for calculating R_j would be more than one and, therefore, a higher value would be obtained that would indicate better accessibility for those villages. The villages beyond the distance of 9 km which do not fall in the catchment area [service area polygons] are considered as inaccessible areas. The value of R_j has been calculated for all the 18 healthcare centres in the study area by Eq. 14.3, in which the reciprocal of R_j values are calculated for each village individually; then the sum of the reciprocal R_j values of the individual villages in each catchment area is calculated as shown in Fig. 14.4a, b, and Table 14.1 (Fig. 14.3).

14.4.3 Accessibility Index (A_i)

In the second step, the accessibility index A_i of the healthcare facilities have been measured by summing up of R_j falling within the catchment area, i.e., if two healthcare centres are in one service area or catchment area of a healthcare centre, the

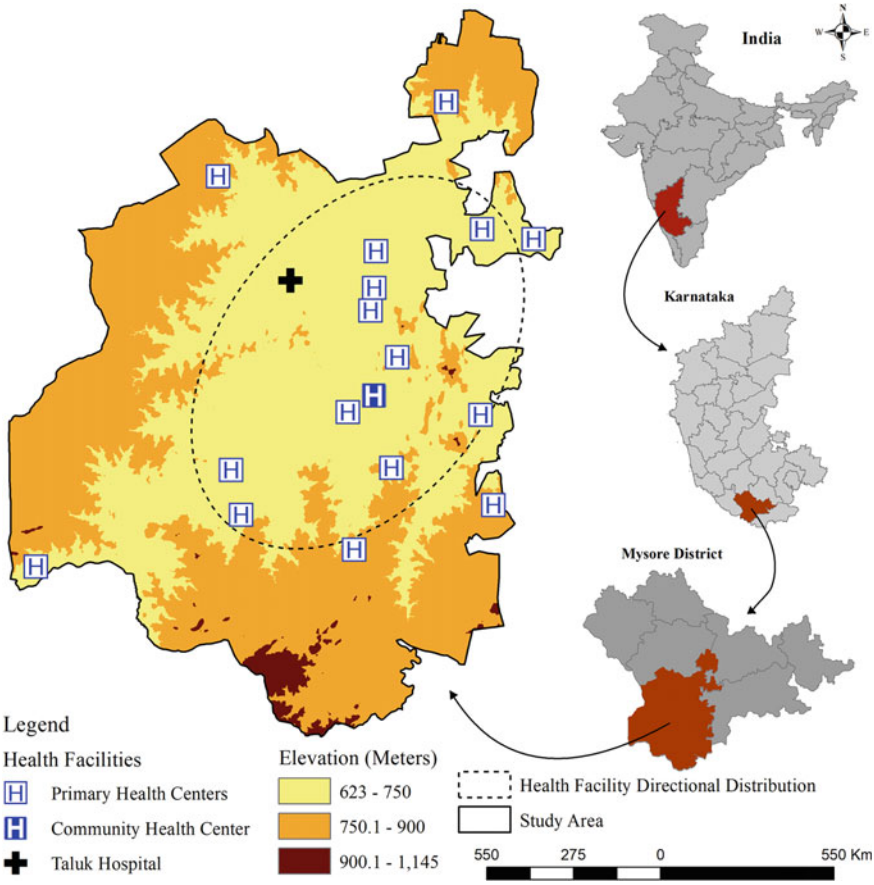

Fig. 14.1 Location map of the study area showing location of major healthcare centres

accessibility index will be the sum of two R_j of that catchment area, Figs. 14.5a, b, and 14.6, Table 14.2.

The maximum and minimum accessibility index of 243 villages is 0.125 and 0 respectively. The mean and standard deviation was 0.726 and 0.010695. As it is evident from the analysis that the healthcare facilities are clustered and unplanned in nature. Due to such distributional patterns many villages are being served by more than one healthcare facility but on the other hand, around 20% of the population of the study area is under-served and therefore doesn't have accessibility to the healthcare system. The main reason for low accessibility is the unplanned and inequitable distribution of healthcare facilities rather than the deficiency of facilities that need to be addressed in healthcare planning (Zhan et al. 2011). The uniform distribution of healthcare facilities can achieve two planning targets: (a) containment of costs by reducing oversupply and (b) equal access by increasing supply in under-served areas (Yang et al. 2006).

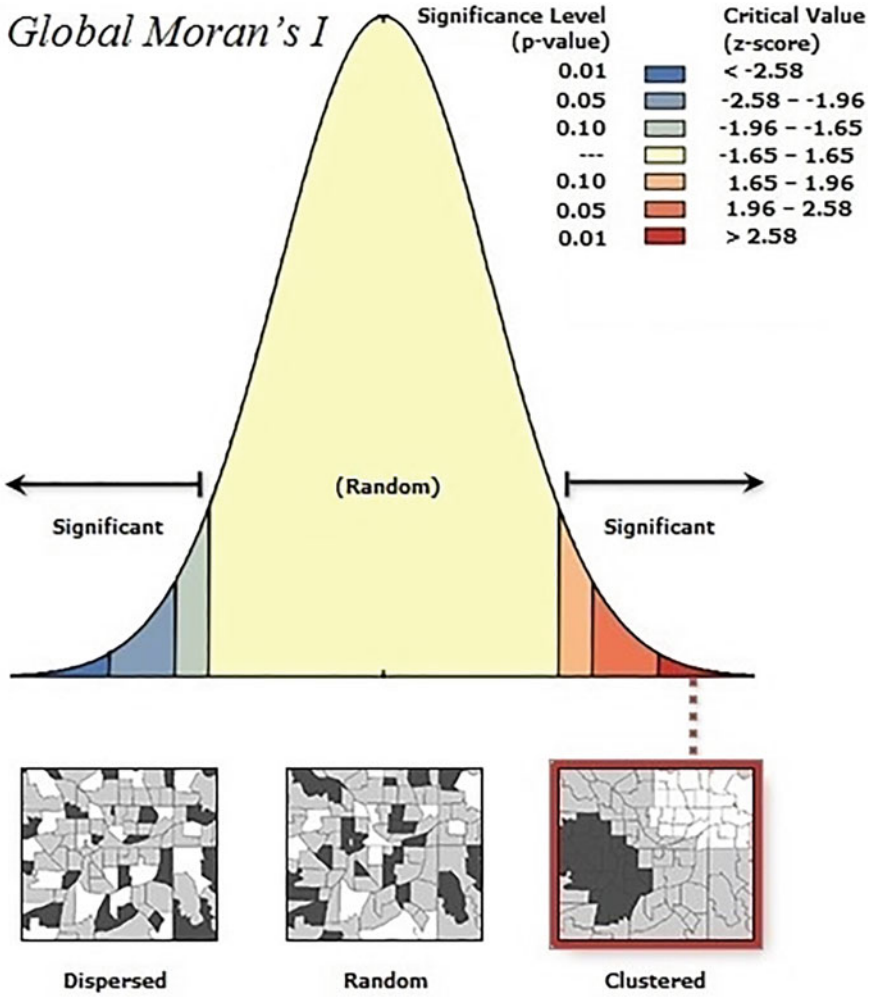

Fig. 14.2 Spatial distribution of healthcare centres based on global Moran's-I

14.4.4 Socio-economic Determinants of Accessibility and Utilization

The patient-exit survey was conducted to measure the socio-economic determinants of accessibility and utilization of healthcare facilities where all the healthcare centres were surveyed. The survey was conducted in order to achieve the objectives under the following headings.

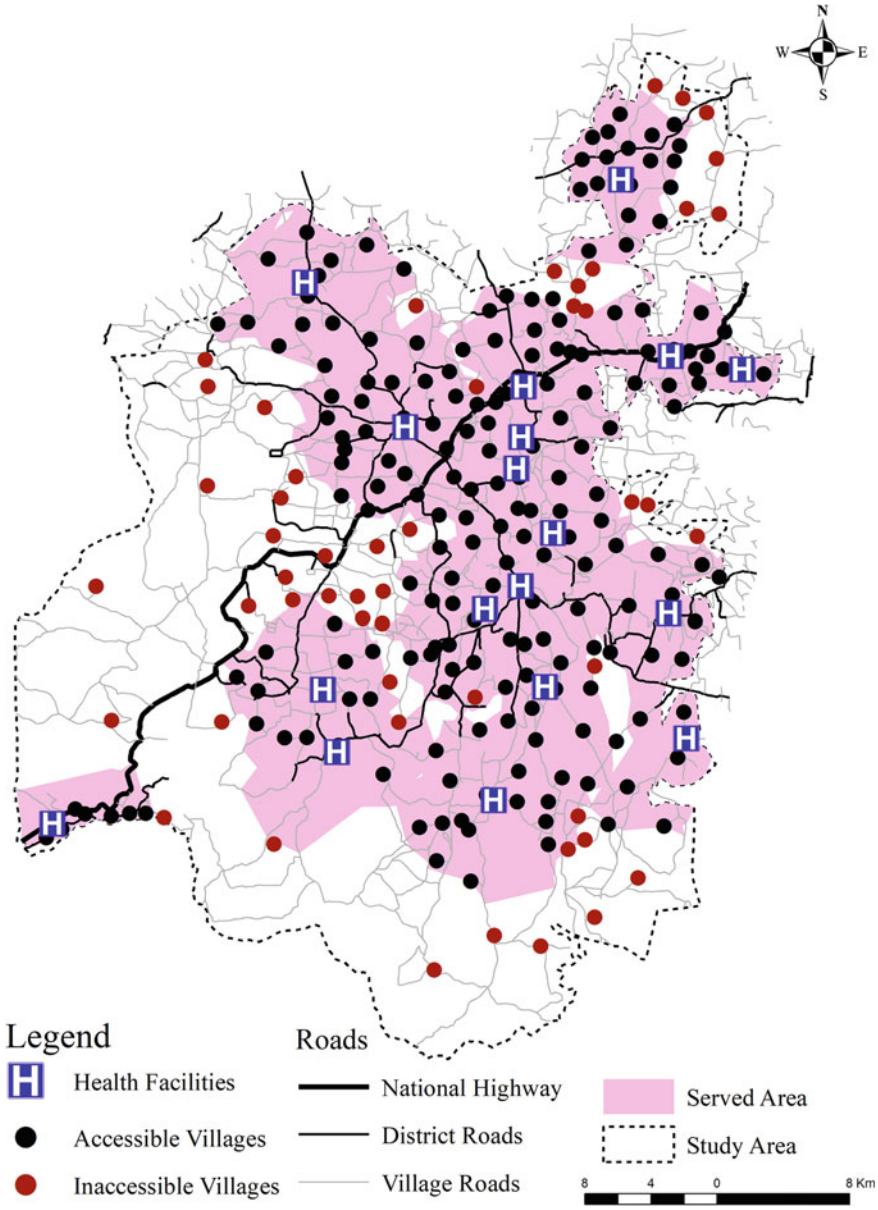

Fig. 14.3 Service area of healthcare centres in the study area showing served and unserved areas

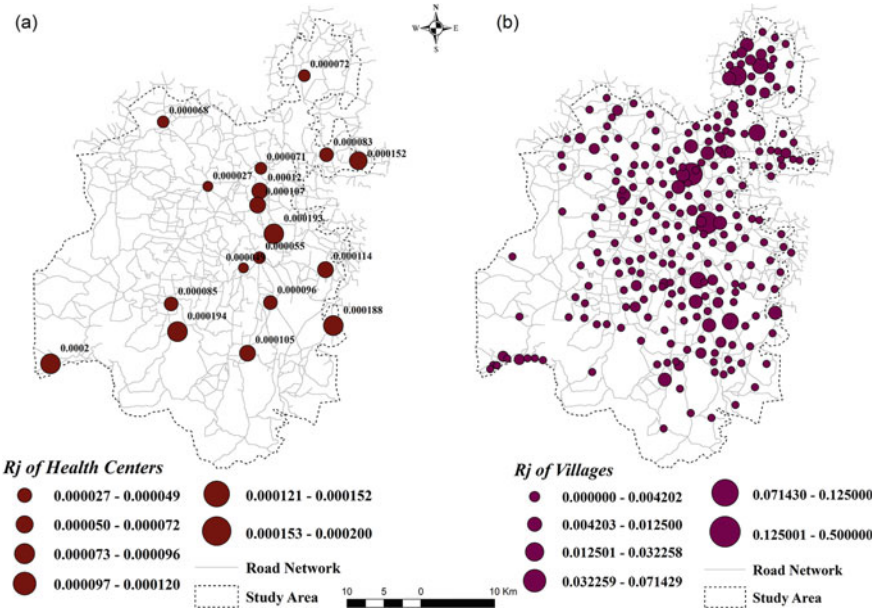

Fig. 14.4 **a** Healthcare centre to population ratio (R_j) of Healthcare centres **b** Healthcare centre to population ratio (R_j) of village

Table 14.1 Healthcare care to population ratio (R_j)

Health centers	Ratio (R_j)	Health centers	Ratio (R_j)
1	7.17463E-05	10	4.92813E-05
2	0.000151883	11	9.55399E-05
3	8.30864E-05	12	8.53971E-05
4	7.11925E-05	13	0.000194269
5	0.00011982	14	0.000105291
6	0.000106933	15	0.00020004
7	6.80921E-05	16	0.000187723
8	0.000192976	17	5.51301E-05
9	0.000114443	18	2.72079E-05

14.4.4.1 Type of Healthcare Use

The private sector in relation to the public healthcare system has played a contrasting role and continues to address healthcare provisions, which have evolved over the period of time and is waving towards the privatization of the healthcare system. Affordability, as well as the accessibility to health centres, makes the patients choose the type of healthcare. The choice of healthcare describes whether the respondents prefer to go for public or private healthcare for their treatment. The majority of the

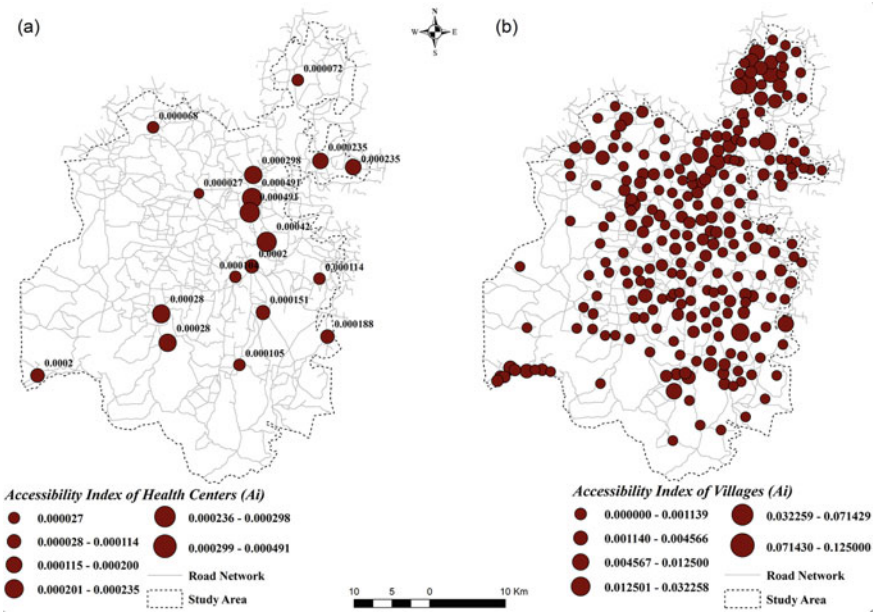

Fig. 14.5 **a** Accessibility index (A_i) of healthcare centres b Accessibility index (A_i) of villages

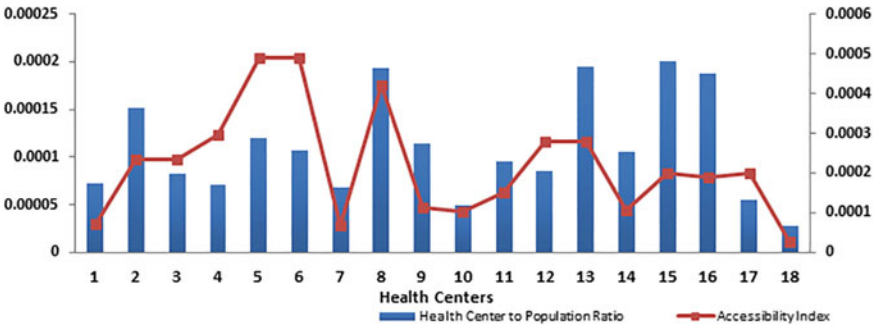

Fig. 14.6 Graph showing relationship between healthcare to population ratio and accessibility index

respondents (85%) in the study area prefer to visit public health centres for their treatment. Whereas, around 15% of the respondent population opted out private healthcare system for their treatment. There are different constraints which can be attributed to the socio-economic conditions of the respondents and the availability of the inadequate quality of healthcare services. In the context of utilization of public healthcare centres and the attitude towards privatization, most of the respondents mentioned that the public healthcare centres are not optimally utilized. Such a situation mainly occurs at a time of disequilibrium between the supply side and the demand side.

Table 14.2 Accessibility index (A_i)

Health centres	Accessibility index (A_i)	Health centres	Accessibility index (A_i)
1	7.17463E-05	10	0.000104411
2	0.000234969	11	0.00015067
3	0.000234969	12	0.000279666
4	0.000297946	13	0.000279666
5	0.000490922	14	0.000105291
6	0.000490922	15	0.00020004
7	6.80921E-05	16	0.000187723
8	0.000419729	17	0.000199951
9	0.000114443	18	2.72079E-05

From the supply side, the primary healthcare centres, as well as higher-order healthcare centres, are not well equipped with the facilities, and from the demand side, the people of the area responded that they prefer to visit higher-order healthcare facilities for better treatment which makes these higher-order health centres overcrowded. Due to these reasons, respondents prefer to visit private clinics/health centres for their treatment. Some of the respondents believe that the income levels, nature of health issues, status in the society, nature of job/occupation, distance to the concerned healthcare centre etc. are also responsible for such reasons. Some local people and doctors working in the study area believe though the present scenario of privatization is growing at a fast rate though it is harmful for the present generation, the people of the area are bound to seek the benefits from the available local healthcare facilities and if the process continues it will become the concern in future.

14.4.4.2 Medicinal Cost

Medical costs indicate the affordability of patients to buy the medicine and other related equipment prescribed by the doctors which are not freely available at the health centres. Arranging money for medicine is directly related to the economic background of the patients. The data depicts the different levels of affordability of the medicinal cost of respondents. The majority of the respondents 74 (37%) are not facing any difficulty in arranging the money for such purposes, 29.5% of the respondents are having marginal difficulty with the arrangement of money for medicines and treatment. About 38 respondents (19%) are having difficulty to afford medicine costs whereas, 14.5% of the respondents are facing it very difficult to arrange money to buy medicines, life-saving drugs or medical equipment needed for their treatment.

14.4.4.3 Accessibility of Patients (Distance)

This represents the distance travelled by patients to reach the nearest healthcare centres for availing services. Transportation is an important factor that influences the accessibility to healthcare facilities mainly in rural areas where people have to travel miles to reach the facility. Transportation distance is considered as a barrier to healthcare access. The data reveals that the majority of the respondents (34%) have access to health care centres within the travel distance of 3 km whereas, 27.5% avail such facilities within the travel distance of 3–6 km. About 21.5% of the respondents have to travel nearly 6–9 km to reach the facility and about 17% have to travel more than 9 km to access healthcare facilities in the study area. It shows the majority of the respondents have access to the nearest PHC facility within the travel distance of 6 km which reflects as having good accessibility to the nearest PHC as prescribed by the National Health Mission (NHM) norms.

14.4.4.4 Travel Time of Patients (Time)

Transportation access based on travel time between human habitations and healthcare services in rural settings is one of the factors which influence the accessibility of the healthcare centre. However, on analysing the data it is revealed that there is a direct access to transportation for the utilization of the rural healthcare system. The travel access to healthcare centres is not only the function of distance but also of conditions of the road, transport frequency and efficiency and topography of the concerned area. The data reveals that the majority of the respondents (40%) have access to PHC's within the travel time of 12 min whereas, 30% of the respondents reach the nearest PHC within the travel time of 12–24 min. 17.5% of the respondents have to travel nearly 24–36 min to reach their nearest PHC and the remaining 12.5% have to travel more than 36 min to access the nearest primary healthcare in the study area.

14.4.4.5 Travelling Cost (in Rupees ₹)

This represents the money spent to reach a particular health centre. Transportation influences healthcare utilization to a great extent mainly in rural areas where patients have to travel more distance to reach their nearest healthcare centre. It has been observed that the transportation cost and the utilization of health centres have negative associations. The travelling cost spent by the respondents to reach the healthcare facility for seeking healthcare facilities reveals that 35.5% of the respondents reach the facility within the travel cost of ₹20 from their respective villages/locations and about 30% spent ₹20–30, whereas 20% of the respondents in the study area have to spend about ₹30–40 to avail the travel facility to reach their nearest healthcare centre and remaining 14.5% of the respondents have to spend more than ₹40 to reach the nearest facility for their treatment from their respective villages.

14.4.4.6 Mode of Transport

In the study area, the respondents were asked about the different modes of transport which they avail to seek the health benefits from their nearest respective health centres. The data reveals that the majority of the respondents in the study area opt for public transport (49%) to access health facilities. They need to travel long distances in order to have access to specialized healthcare services. This long distance travel has become a significant burden in terms of travel time; the lack of reliable means of transport, travel costs, and time are the main obstacles to healthcare accessibility.

14.4.4.7 Waiting Time for Transportation

Waiting time is one of the constraints of the accessibility to healthcare centres which in turn affects the utilization of healthcare services. Due to the less efficient transportation services patients have to wait to avail and access healthcare facilities. Data reveals that the majority of the respondents (52.5%) have to wait more than 10 min to get the transport to reach the healthcare centre. About 31% of the respondents have to wait about 10–20 min, whereas 16.5% have to wait more than 20 min to get the transport to reach the nearest healthcare centre.

14.4.5 Quality of Healthcare Services

The utilization of services by the user population and their perception is the key for determining the quality of a particular service and not the one that is defined by medical professionals. When users perceive the quality of a particular service or facility to be high, they will be motivated to seek care for illnesses. Similarly, when the population perceives the quality of healthcare provided by a given facility to be below standards, they will seek alternative sources of healthcare. Quality of healthcare services includes the 19 types of quality services that were asked to the respondents to analyse the quality of healthcare facilities available in the health care centres and they were asked to give a rating between poor, fair, good, and excellent. Under the quality of services, healthcare delivery is very important and it includes the variables such as adequate availability of doctors and other technical human resources, services provided at healthcare centres like X-ray, USG, blood tests, etc., satisfaction over the prescriptions, good diagnosis, quality of drugs and sufficient time for analysis/diagnosis of patients. Here the majority of the respondents rated excellent condition and the least responded as poor condition. Under the heading of the availability of doctors, 52.6% of the respondents responded the availability as excellent and only 5% responded as poor whereas, good and fair availability of doctors was responded by 31.4% and 11% of the respondents respectively. The diagnosis/examination by the doctors was rated as excellent by 41.6% of the respondents and only 6.4% responded it under the poor category. Whereas, good and fair

for the good diagnosis were responded by 43% and 9% of the respondents respectively. In the satisfaction with the prescriptions prescribed by the doctors, 55% of the respondents responded excellent and only 4.6% of the respondents termed it as poor. Whereas, good and fair responses were responded by 37% and 3.4% of the respondents respectively. To examine the quality of drugs available in health centres, the respondents rated 42.4% as excellent and about 1.8% responded as poor. Whereas, good and fair for the quality of drugs were responded by 42.6 and 13.2% of the respondents respectively. In the response asked for sufficient time spent by doctors with the patients, 46.6% of the respondents responded as excellent and only 5.6% responded as poor. Whereas, good and fair responses were responded by 44.4 and 3.4% of the respondents respectively.

Overall reception facility, follow-up of patients, good clinical examination, monitoring of patients, and availability of adequate medical equipment in the health centres come under the interpersonal and diagnostic aspect of care. For the above mentioned criterion, respondents were asked about the overall reception facility, seating facility, cleanliness, etc., for which around 54%of the respondents responded as excellent and only 2.8% responded as poor. While as, good and fair responses were responded by 39.2 and 4% of the respondents respectively. About the good clinical examination, 41.6% of the respondents were opinioned as excellent and 3.6% response was poor. The patient's follow-up and monitoring of the visit of the patients by the doctors are very good and the majority of the respondents (53.2%) said it is excellent. The majority of the respondents (40%) opined that the medical equipment is adequately available.

The respondents opined that the facilities include the adequacy of rooms for a check-up, adequate availability of doctors for women separately, neat and clean hospital premises and proper disposal of medical waste are excellent, and very few respondents reported that the facilities as poor. The health personal conduct and drug availability include the support and compassion of doctors towards the patients, respect for the patients by the health personnel and availability of all the drugs prescribed by the doctors in the health centres. The respondents believe that all these facilities are excellent for the patients for availing the healthcare facility. Financial and physical access to healthcare facilities includes financial feasibility of treatment of the patient, ease of obtaining drugs, and easy approachability and the respondents opine that all of these are excellent to the patients for availing the healthcare facilities.

14.4.6 Behaviour Aspects

There has been considerable discussion on the behavioural aspects of healthcare personnel and their patients with a focus on the utilization of facilities at healthcare centres. In addition to the socio-economic determinants, the behaviour of the healthcare staff including doctors, nurses, paramedical staff, ambulatory services, sanitation, electricity, cleanliness, etc. are some of the determinants which have greatly marked their footprints on the utilization of healthcare centres. The utilization of

healthcare facilities is determined to a great extent by the behaviour of healthcare staff. As a result, a community's attention-seeking behaviour influences how healthcare facilities are used, and therefore the population's health outcomes. Respondent's perception of doctor's relationship towards patients was also examined. About 43.8% of the respondents perceived that the communication of doctors is good with the patients; 32.6% of them have perceived fair behaviour of doctors. However, a low percentage of people (13.6%) perceived the poor behaviour of doctors towards the patients. About 48.6% of the respondents believe that the behaviour of nurses in the healthcare centres is good, whereas, 33.2% perceived that they are fair in behavioural aspects and 15.2% opined that the attitude of nurses towards the patient is excellent. Paramedical staff in healthcare centres includes lab assistants/technicians, group D workers, and clerks. About 56% of the respondents have mentioned that their way of behaviour is good, whereas, 1.2% perceived that their behaviour is poor with them during the treatment. The behavioural aspect of other supporting staff has also been observed fairly, as more than 50% of the respondents believe that their behaviour has been observed well.

14.5 Discussion

Accessibility to healthcare services is largely determined by the distribution of health centres, population distribution and transport facilities (Delamater et al. 2012; Das et al. 2020). Accessibility disparity in healthcare relies on many variables, so the existence of the variables driving access differences must be explored (Wang and Luo 2005). The three key analytical concerns on which the creation and scope of the healthcare centres can be delineated are the availability of healthcare facilities, location characteristics, and level of access to healthcare facilities (Murad 2011). In most healthcare systems, equity in the provision of healthcare facilities and fair access to these facilities has become a major concept (Mackenbach et al. 2008). It is, therefore, necessary to determine the nature of the spatial conditions and the estimation of the healthcare facilities needed for a given population, which is indeed important for the proper utilization of healthcare facilities, and there are many techniques by which the spatial distribution of healthcare facilities can be measured, such as the location quotient, Gini's Coefficient, Lorenz curve, Average Nearest Neighbour, etc. (Tali et al. 2017a). The study area has varied and undulating terrain, slope and steepness, physical characteristics in terms of location and design of healthcare centres, road density, road form, road conditions, etc. have had a significant effect on accessibility. The study indicates that the majority of residents in the study area opt for public transport to access healthcare facilities. They have to travel long distances to access subspecialty healthcare services. This long distance travel has become a significant burden in terms of travel time, the lack of reliable means of transport, travel costs and time are the main obstacles to accessing the healthcare systems. Health education is also another barrier to accessing healthcare facilities in rural societies as compared to their urban counterparts. Poor knowledge regarding health affects the patient's ability

to understand health information and the instructions of their health care providers. The literacy rate of the study area is 64.11% even then a large section of the residents are reluctant to see a doctor due to fear or frustration from communicating with a healthcare professional for that reason sometimes they even consult local medical shopkeepers to get their treatment done.

The study revealed that the access and utilisation of public healthcare facilities have been greatly influenced by a number of obstacles. Individuals without health insurance have less access to healthcare systems. Data from the survey indicate that majority of the patients (95%) do not have insurance coverage data. The data also shows that 5% of the population protected by insurance is from the Municipality limits of the study area. The problem of social stigma and privacy is another obstacle to access healthcare services, particularly in rural areas. Patients were asked if such chronic diseases were freely discussed with the physicians. 84% of patients do not want to reveal their concerns regarding mental health, drug abuse, sexual health, pregnancy or even common chronic disease due to social stigma and privacy issues. Healthcare workers and infrastructure shortages have a major impact on the use of healthcare facilities. While the population healthcare ratio in the sample is below the NHM levels, the survey showed that (35.6%) there was a lack of availability of healthcare providers and infrastructure facilities. In most healthcare centres, doctors often attend late-hour work, and some patients point out that some healthcare centres do not even have basic infrastructure facilities. All these findings suggest that social accessibility has a significant role to play in reaching out and using healthcare services.

14.6 Conclusion

The two-step floating catchment area (2SFCA) was used to calculate the accessibility of villages in this article. It is centred on the potential model and incorporates the interaction between supply and potential demand. The results of the 2SFCA analysis and mapping demonstrate that several improvements can be made to study the spatial relationships between facility seekers in the form of population and providers to monitor the accessibility scenario of the area under study. The 2SFCA approach is the best possible mechanism for developing a better estimate of spatial accessibility. Furthermore, the idea of modifying the 2SFCA method to answer additional questions using current GIS technology provides benefits to analyse the spatial relations for decision-makers to improve the healthcare system of an area. The socio-economic part of the study reveals that the health status and access to healthcare of rural populations indicate that rural populations of Heggadadevanakote (HD Kote) have less access to such facilities based on the dimensions of accessibility, proximity, and quality, compared to their urban counterparts. Travelling to a primary care provider can be expensive and cumbersome for patients living in remote rural areas, mainly in the south and southeast area of the taluk, since subspecialty care (District Hospital) is also far away (56 km). Most of the patients replace local primary care providers with

subspecialists or decide to postpone or suspend care. The majority of the developed areas in the world lack a sustainable health care policy, the study hold significance by aiding policymakers to distinguish the accessibility of healthcare centres in the study area along with the unserved population so that a robust health care system can be put to use keeping in mind the location aspects.

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References

Apparicio P, Abdelmajid M, Riva M, Shearmur R (2008) Comparing alternative approaches to measuring the geographical accessibility of urban health services: distance types and aggregation-error issues. Int J Health Geogr 7(1):1–14

Brabyn L, Gower P (2003) Mapping accessibility to general practitioners. In: Geographic information systems and health applications (pp 290–308)

Bryant J Jr, Delamater PL (2019) Examination of spatial accessibility at micro-and macro-levels using the enhanced two-step floating catchment area (E2SFCA) method. Ann GIS 25(3):219–229

Das LT, Kaushal R, Garrison K, Carrillo V, Grinspan Z, Theis R, Abramson E (2020) Drivers of preventable high health care utilization: a qualitative study of patient, physician and health system leader perspectives. J Health Serv Res Policy 25(4):220–228

Delamater PL, Messina JP, Shortridge AM, Grady SC (2012) Measuring geographic access to health care: raster and network-based methods. Int J Health Geogr 11(1):1–18

Fayaz A, Ul Shafiq M, Singh H, Ahmed P (2020) Assessment of spatiotemporal changes in land use/land cover of North Kashmir Himalayas from 1992 to 2018. Model Earth Syst Environ 1–12

Halder B, Bandyopadhyay J, Banik P (2020) Assessment of hospital sites' suitability by spatial information technologies using AHP and GIS-based multi-criteria approach of Rajpur-Sonarpur municipality. Model Earth Syst Environ 6(4):2581–2596

Higgs G, Gould M (2001) Is there a role for GIS in the 'new NHS'? Health Place 7(3):247–259

Horev T, Pesis I, Mukamel DB (2004) Trends in geographic disparities in allocation of health care resources in the US. Health Policy 68(2):223–232

Kanuganti S, Sarkar AK, Singh AP (2016) Quantifying accessibility to health care using Two-step Floating Catchment Area Method (2SFCA): a case study in Rajasthan. Transportation Research Procedia 17:391–399

Luo W (2004) Using a GIS-based floating catchment method to assess areas with shortage of physicians. Health Place 10(1):1–11

Luo W, Qi Y (2009) An enhanced two-step floating catchment area (E2SFCA) method for measuring spatial accessibility to primary care physicians. Health Place 15(4):1100–1107

Luo W, Wang F (2003) Measures of spatial accessibility to health care in a GIS environment: synthesis and a case study in the Chicago region. Environ Plann B Plann Des 30(6):865–884

Mackenbach JP, Stirbu I, Roskam AJR, Schaap MM, Menvielle G, Leinsalu M, Kunst AE (2008) Socioeconomic inequalities in health in 22 European countries. N Engl J Med 358(23):2468–2481

Mansour S (2016) Spatial analysis of public health facilities in Riyadh Governorate, Saudi Arabia: a GIS-based study to assess geographic variations of service provision and accessibility. Geo-Spatial Inf Sci 19(1):26–38

McGrail MR (2012) Spatial accessibility of primary health care utilising the two step floating catchment area method: an assessment of recent improvements. Int J Health Geogr 11(1):1–12

McGrail MR, Humphreys JS (2009) Measuring spatial accessibility to primary care in rural areas: improving the effectiveness of the two-step floating catchment area method. Appl Geogr 29(4):533–541

Minutha V, Subash SS, Tali JA, Divya S (2014) Measuring spatial accessibility to primary health center in Mysore district. Int J Phys Soc Sci 4(12):30

Murad AA (2011) Creating a geographical information systems-based spatial profile for exploring health services supply and demand. Am J Appl Sci 8(6):644

Ngui AN, Apparicio P (2011) Optimizing the two-step floating catchment area method for measuring spatial accessibility to medical clinics in Montreal. BMC Health Serv Res 11(1):1–12

Pramanik MK (2016) Site suitability analysis for agricultural land use of Darjeeling district using AHP and GIS techniques. Model Earth Syst Environ 2(2):56

Radke J, Mu L (2000) Spatial decompositions, modeling and mapping service regions to predict access to social programs. Geogr Inf Sci 6(2):105–112

Rosenblatt RA, Lishner DM (1991) Surplus or shortage? Unravelling the physician supply conundrum. West J Med 154(1):43

Tali AF, Asima N, Aasif N (2017a) Accessibility to higher order public health centers a study of district Pulwama (J&K). Int J Sci Res Sci Technol 3:55–61

Tali JA, Divya S, Nusrath A (2017b) Spatial distribution of health centers: a study of district Pulwama (Jammu and Kashmir)

Wang F, Luo W (2005) Assessing spatial and nonspatial factors for healthcare access: towards an integrated approach to defining health professional shortage areas. Health Place 11(2):131–146

Wong DW (2005) Statistical analysis of geographic information with ArcView GIS and ArcGIS

Yang DH, Goerge R, Mullner R (2006) Comparing GIS-based methods of measuring spatial accessibility to health services. J Med Syst 30(1):23–32

Zhan Q, Wang X, Sliuzas R (2011, June) A GIS-based method to assess the shortage areas of community health service—case study in Wuhan, China. In: 2011 international conference on remote sensing, environment and transportation engineering. IEEE, pp 5654–5657